NATIONAL ATHLETICS ACADEMY INC.

Full Website Technology Stack & Architecture

Astro • Tailwind CSS • TypeScript • PHP • cPanel/Apache • Cloudflare

2.6 Brand Identity

The NAA identity is a circular crest built on a maple leaf, three athlete silhouettes and three stars, framed by a navy ring carrying the wordmark and a gold "CANADA" mark at the base. The crest signals national scope, multi-sport delivery and institutional credibility.

Role	Colour	HEX	RGB	Primary use
Primary	Navy Blue	#021F47	2, 31, 71	Headers, navigation, footers, body text, structural fields
Accent / action	Maroon Red	#83161B	131, 22, 27	Buttons, calls to action, active states, key headings
Secondary accent	Deep Maple Red	#7D161A	125, 22, 26	Hover states, gradients, secondary accents
Highlight	Gold	#DDAA40	221, 170, 64	Badges, awards, achievement markers, small highlights
Base	White / Off-white	#FDFCFD	253, 252, 253	Content backgrounds, reversed text
Neutral	Light Grey	#F2F3F5	242, 243, 245	Cards, table banding, section dividers

- **Accessibility note.** Navy on white and white on navy or maroon meet WCAG 2.1 AA for body text. Gold has insufficient contrast on both white and maroon backgrounds and must be reserved for large headings, icons and decorative accents — never body copy.
- **Usage discipline.** Maroon should be reserved for interactive and action elements so that red does not read as a system warning. Navy carries the structural weight of the interface.

Project direction

A fast, accessible, SEO-friendly, mostly static website for National Athletics Academy Inc. The technical foundation may reuse proven Astro/PHP architecture, but all NAA branding, credentials, analytics IDs, domains, email settings and secrets are newly configured.

1. Executive Technical Overview

NAA will use Astro as the primary frontend framework and static site generator. Tailwind CSS v4 provides the styling system, with TypeScript/Astro components for the UI. PHP is reserved for server-side forms and integrations such as reCAPTCHA verification and email delivery. Production hosting is cPanel-based with Apache, with the generated site deployed into **public_html/**.

Architecture principle: static-first frontend, minimal client-side JavaScript, and private credentials kept server-side.

2. Core Technology Stack

Category	Technology	Purpose
Framework	Astro v6.x	Static-first website framework and component system
Rendering	Static Site Generation (SSG)	Pre-generates public pages as HTML
Build	Vite	Development server and production build
Language	TypeScript + Astro	Typed frontend/component development
Runtime	Node.js ≥ 22.12.0	Local development and build environment
Package manager	npm	Dependencies and scripts
CSS	Tailwind CSS v4	Responsive utility-first styling
Integration	@tailwindcss/vite	Vite-native Tailwind v4 integration
Images	Astro image handling + Sharp	Image optimization and responsive delivery
Client JS	Vanilla JS / TypeScript	Small interactive features
Animation	IntersectionObserver	Lightweight scroll animations
Backend	PHP 8.x	Forms and server-side integrations
CAPTCHA	Google reCAPTCHA Enterprise	Bot/abuse protection
Email	Authenticated SMTP / transactional API	Reliable form notifications
Analytics	Google Tag Manager	Analytics and conversion tracking
Hosting	cPanel + Apache	Production hosting
CDN/DNS	Cloudflare (optional)	DNS, SSL, CDN, WAF/security
Database	None initially	Keep architecture simple
Version control	Git	Source control

3. NAA Brand Identity & Design System

The NAA identity is a circular crest featuring a maple leaf, three athlete silhouettes and three stars, framed by a navy ring carrying the wordmark and a gold “CANADA” mark. It communicates national scope, multi-sport athletics, achievement and institutional credibility.

Role	Colour	HEX	RGB	Primary use
Primary	Navy Blue	#021F47	2, 31, 71	Headers, navigation, footer, body text, structural fields

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Accessibility: navy on white and white on navy/maroon should carry normal text. Gold should be reserved for large headings, icons, badges and achievement accents where contrast is adequate. Maroon is an action colour, not a warning/error colour.

Typography

- **Outfit:** headings, navigation, program names, major statistics and promotional display text.
- **Inter:** paragraphs, forms, descriptions, tables and supporting interface text.

Visual direction

National • Athletic • Professional • Modern • Credible • Canadian. Navy provides structural weight; maroon drives action; maple red supports interaction; gold signifies achievement.

4. Frontend Architecture

The frontend uses reusable Astro components. React is not required unless a future feature genuinely needs a client framework.

```
src/
  components/
    Header.astro
    MobileMenu.astro
    Footer.astro
    Hero.astro
    SectionHeading.astro
    Button.astro
    ProgramCard.astro
    CoachCard.astro
    Programs.astro
    Coaches.astro
    Locations.astro
    FAQ.astro
    Contact.astro
    RegistrationForm.astro
  layouts/
    Layout.astro
  pages/
  styles/
    global.css
  data/
    programs.ts
    coaches.ts
    locations.ts
```

Client-side functionality

- Mobile navigation, FAQ accordions and accessible interactive controls.
- AJAX form submission and client-side validation.

- Lightweight scroll animations with IntersectionObserver.
- Deferred loading of analytics, maps and other third-party scripts.
- Support for prefers-reduced-motion.

5. Website Information Architecture

```
/
■■■ about/
■■■ programs/
■■■ athletics/
■■■ coaches/
■■■ registration/
■■■ locations/
■■■ resources/
■■■ news-events/
■■■ contact/
```

Program-specific pages can be added as NAA expands. Only publish pages representing real Academy offerings.

6. Images, Performance & Core Web Vitals

- Use Astro image handling and Sharp.
- Prefer WebP/AVIF where appropriate.
- Generate responsive image sizes and avoid oversized originals.
- Lazy-load below-the-fold imagery.
- Optimize the hero/LCP image carefully.
- Keep JavaScript and third-party scripts minimal.
- Defer analytics, maps, social embeds and other non-essential scripts.

Target: strong Core Web Vitals and fast first load on mobile networks.

7. SEO & Structured Data

- Unique page titles and meta descriptions.
- Canonical URLs.
- Open Graph and social metadata.
- XML sitemap and robots.txt.
- Semantic HTML and correct heading hierarchy.
- Descriptive image alt text.
- Clean URLs.
- Schema.org structured data where accurate.

Potential structured data: Organization, SportsOrganization, EducationalOrganization, LocalBusiness, Person, Event, Course, WebSite and BreadcrumbList.

8. Contact, Registration & Interest Forms

```
Visitor
  ↓
Astro form
  ↓
Client validation
  ↓
reCAPTCHA Enterprise token
  ↓
POST to PHP
  ↓
Server validation + sanitization
  ↓
reCAPTCHA verification
  ↓
Rate limiting / abuse checks
  ↓
Authenticated email delivery
  ↓
JSON success/error response
```

Recommended reCAPTCHA actions

```
CONTACT
PROGRAM_INTEREST
ATHLETE_REGISTER
COACH_REGISTER
VOLUNTEER_INTEREST
GENERAL_INQUIRY
```

9. Security Architecture

- HTTPS enforced in production.
- Google reCAPTCHA Enterprise on public forms.
- Server-side validation regardless of frontend validation.
- Input sanitization and email header-injection protection.
- Rate limiting for public form endpoints.
- CSRF protection where sensitive state-changing actions require it.
- Private secrets kept outside Git and outside the public web root.
- Cloudflare WAF/bot controls may be enabled.
- Production errors logged server-side rather than exposed to visitors.

No Wascana FC credentials, domains, API keys, reCAPTCHA keys, Google Cloud project IDs, GTM IDs, SMTP passwords, Cloudflare tokens or other secrets are to be reused.

10. Google reCAPTCHA Enterprise

Create a new Google Cloud project and new reCAPTCHA Enterprise configuration for NAA. The browser-visible site key is separate from private server credentials.

```
# Example only — never commit real production values
PUBLIC_RECAPTCHA_SITE_KEY=
RECAPTCHA_PROJECT_ID=
RECAPTCHA_API_KEY=
```

The server should verify token validity, expected action and a configured score threshold. Tune the threshold using real NAA traffic rather than copying another organization's setting.

11. Email & Notification Architecture

Use authenticated SMTP or a transactional email API for reliable production delivery. cPanel-hosted mailboxes can be used for NAA organizational addresses.

```
Website
  ↓
PHP backend
  ↓
Authenticated SMTP / transactional email service
  ↓
NAA mailbox
```

12. Analytics & Tracking

Use a new Google Tag Manager container owned/configured for NAA. Never reuse another organization's GTM container.

- program_view
- registration_start
- registration_submit
- contact_submit
- coach_application
- phone_click
- email_click
- location_click

Load analytics and other non-essential scripts in a deferred manner where practical.

13. Production Hosting — cPanel

Layer	Production choice
Hosting control panel	cPanel
Web server	Apache
Frontend	Astro-generated static files
Backend	PHP 8.x, subject to hosting provider support
Production root	public_html/
SSL	cPanel AutoSSL / Let's Encrypt, depending on host
DNS/CDN	Cloudflare optional
Database	None initially
Node.js in production	Not required for static deployment

Deployment model

```
Local development
      ↓
npm run build
      ↓
Astro generates dist/
```

```
      ↓
Upload contents of dist/ to cPanel public_html/
      ↓
Apache serves static HTML/CSS/JS/assets
      ↓
PHP files run server-side for forms
```

Astro does not need to run as a Node.js server in production for this architecture.

14. cPanel Services & Configuration

- File Manager for deployment and maintenance.
- Domains / Subdomains.
- SSL/TLS and AutoSSL.
- PHP version selector / MultiPHP Manager.
- Email Accounts and Email Routing.
- Cron Jobs only when future scheduled server tasks are required.
- Error Logs for troubleshooting.
- Redirects for controlled URL changes.
- Git Version Control if enabled by the host.
- MySQL/MariaDB only if a future feature genuinely needs persistent data.

Recommended public_html layout

```
public_html/
■■■ index.html
■■■ about/
■■■ programs/
■■■ coaches/
■■■ registration/
■■■ contact/
■■■ assets/
■■■ images/
■■■ contact.php
■■■ registration.php
■■■ coach-register.php
■■■ interest.php
■■■ verify-recaptcha.php
■■■ robots.txt
■■■ .htaccess
```

15. Cloudflare

Cloudflare is optional but recommended when the hosting/DNS arrangement supports it. It can provide DNS, TLS, CDN caching, WAF and additional bot/security controls.

```
Visitor
  ↓
Cloudflare DNS / CDN / WAF
  ↓
cPanel / Apache
  ↓
public_html/
```

All Cloudflare accounts, tokens and configuration must belong to NAA.

16. Environment Variables & Secrets

Separate development and production configuration. Never commit real secrets to GitHub or put private credentials in browser-delivered JavaScript.

```
# .env.example
PUBLIC_SITE_URL=
PUBLIC_RECAPTCHA_SITE_KEY=
RECAPTCHA_PROJECT_ID=
RECAPTCHA_API_KEY=
PUBLIC_GTM_ID=
CONTACT_EMAIL=
SMTP_HOST=
SMTP_PORT=
SMTP_USERNAME=
SMTP_PASSWORD=

.gitignore
.env
.env.*
!.env.example
node_modules/
dist/
```

On cPanel, private PHP configuration should preferably live outside public_html/ or be protected by the hosting environment. Never expose API keys or SMTP passwords through public files.

17. Recommended Project Structure

```
national-athletics-academy/
■
■■■ public/
■   ■■■ images/
■   ■■■ icons/
■   ■■■ favicon.svg
■   ■■■ contact.php
■   ■■■ registration.php
■   ■■■ coach-register.php
■   ■■■ interest.php
■   ■■■ verify-recaptcha.php
■   ■■■ robots.txt
■   ■■■ .htaccess
■
■■■ src/
■   ■■■ components/
■   ■■■ layouts/
■   ■   ■■■ Layout.astro
■   ■■■ pages/
■   ■■■ styles/
■   ■   ■■■ global.css
■   ■■■ data/
■       ■■■ programs.ts
■       ■■■ coaches.ts
■       ■■■ locations.ts
■
■■■ .env.example
■■■ .gitignore
■■■ astro.config.mjs
■■■ package.json
■■■ tsconfig.json
■■■ README.md
```

18. Accessibility

Target: WCAG 2.1 AA.

- Keyboard-accessible navigation and controls.
- Visible focus indicators.
- Sufficient text/background contrast.
- Proper form labels and error messages.
- Semantic landmarks and headings.
- Descriptive alt text for meaningful images.
- Reduced-motion support.
- Touch-friendly interactive targets.

19. Testing & Quality Assurance

Area	Checks
Functionality	Navigation, mobile menu, forms, registration, email delivery, CAPTCHA, links, FAQ, maps
Browsers	Chrome, Edge, Firefox, Safari, iOS Safari, Android Chrome
Responsive	Mobile, tablet, laptop, desktop and large desktop
SEO	Titles, metadata, canonical URLs, sitemap, robots, schema, social previews
Security	Injection, rate limits, invalid CAPTCHA, missing fields, oversized/malicious input
Accessibility	Keyboard use, focus, labels, contrast, semantics, reduced motion
Performance	Core Web Vitals, image sizes, JS payload, third-party script impact

20. Development & Deployment Workflow

```
1. Develop locally in VS Code / Laragon.
2. Install dependencies with npm.
3. Run Astro development server.
4. Test frontend and PHP endpoints locally.
5. Run: npm run build
6. Review dist/.
7. Upload generated files to cPanel public_html/.
8. Configure production PHP and private credentials.
9. Configure DNS/SSL/Cloudflare as applicable.
10. Test live forms and email delivery.
11. Verify analytics, SEO, sitemap and robots.txt.
12. Perform final mobile/accessibility/security checks.
```

21. Separation From the Existing Wascana Project

Reuse technical patterns where useful, but treat NAA as a completely independent organization and production environment.

Remove or replace all Wascana-specific items

- Organization name, abbreviations, domain, email, phone and address.
- Logos, photographs, graphics, copy, programs, registration links, coach information and social links.

- Google reCAPTCHA site keys, API keys and Cloud project IDs.
- Google Tag Manager container IDs.
- SMTP credentials and email configuration.
- Cloudflare credentials, tokens and account configuration.
- Private keys, passwords, secrets, authentication tokens and session secrets.

Repository-wide audit search

```
wascana
Wascana
wascanafc
GTM-
recaptcha
api_key
site_key
project_id
secret
token
password
smtp
```

22. Database Strategy

No database is required for the initial public website if programs, coaches and locations can be maintained as Astro data/content. Add MySQL/MariaDB later only if NAA needs authenticated users, complex registration, payment workflows, dashboards or other dynamic features.

```
Initial:
Astro content/data → Static pages

Future, if required:
Database → PHP/API → Dynamic application features
```

23. Production Launch Checklist

- NAA domain configured and HTTPS working.
- cPanel document root points to public_html/.
- Astro production build uploaded successfully.
- PHP version selected and tested.
- NAA reCAPTCHA Enterprise configured.
- Private API credentials are not publicly accessible.
- SMTP/email delivery tested.
- Form rate limiting tested.
- Cloudflare configuration verified, if used.
- GTM uses a new NAA container.
- Sitemap and robots.txt available.
- Canonical and social metadata verified.
- Favicon and NAA crest installed.

- NAA brand colours and typography applied consistently.
- No Wascana-specific credentials, content or identifiers remain.
- Mobile, browser, accessibility, security and performance testing completed.

24. Final Technology Stack — Quick Reference

Category	NAA implementation
Framework	Astro v6.x
Rendering	Static Site Generation (SSG)
Build	Vite
Language	TypeScript / Astro
Node.js	≥ 22.12.0 for local development/build
CSS	Tailwind CSS v4 + @tailwindcss/vite
Fonts	Outfit + Inter
UI	Astro components
Client JS	Vanilla JS / TypeScript
Animation	IntersectionObserver
Images	Astro image handling + Sharp
Forms	Astro frontend + PHP backend
CAPTCHA	Google reCAPTCHA Enterprise
Email	Authenticated SMTP / transactional email API
Analytics	Google Tag Manager — new NAA container
SEO	Metadata + sitemap + Schema.org + canonical URLs
Security	HTTPS + CAPTCHA + validation + rate limiting + optional Cloudflare WAF
Hosting	cPanel
Web server	Apache
Production root	public_html/
PHP	PHP 8.x, subject to host support
SSL	cPanel AutoSSL / Let's Encrypt
CDN/DNS	Cloudflare optional
Database	None initially
Local development	VS Code + Node.js + Laragon
Version control	Git
Deployment	Local Astro build → cPanel public_html/
Accessibility	WCAG 2.1 AA target
Brand	NAA Navy / Maroon / Deep Maple Red / Gold / Off-white / Light Grey

Final architecture: Astro + Tailwind CSS v4 + TypeScript for the frontend; PHP for server-side forms; cPanel/Apache for production hosting; optional Cloudflare for DNS/CDN/security; Google reCAPTCHA Enterprise for form protection; authenticated email for notifications; and a completely new NAA-specific credential and analytics environment.